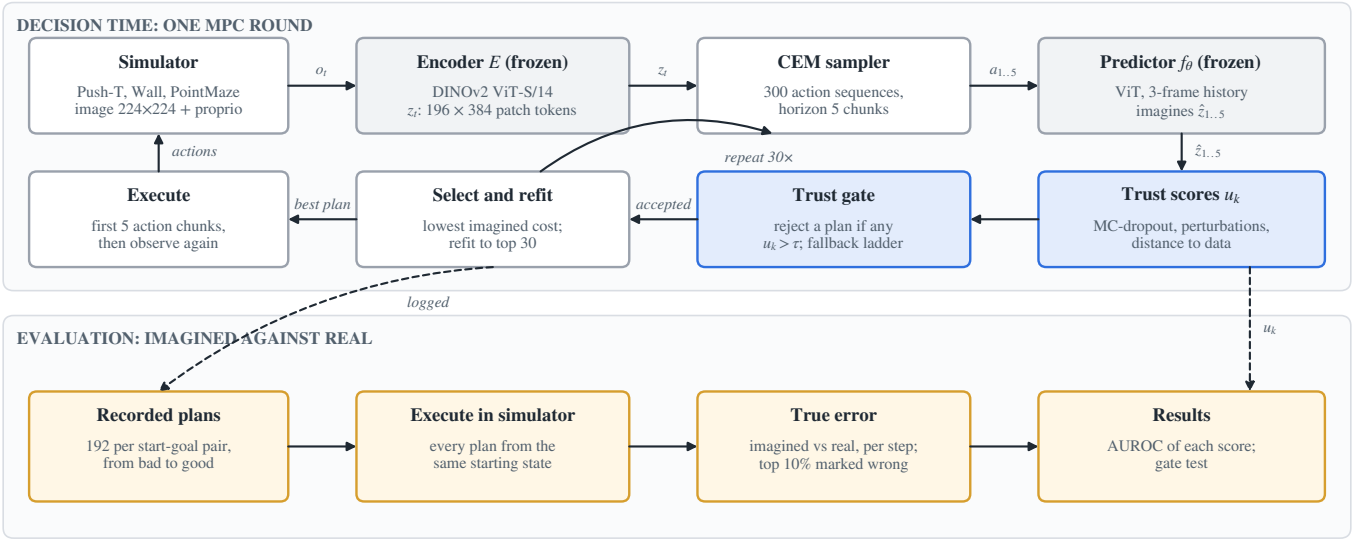
DECISION TIME: ONE MPC ROUND
Simulator
Push-T, Wall, PointMaze
image 224×224 + proprio
o_t
Encoder E (frozen)
DINOv2 ViT-S/14
z_t: 196 × 384 patch tokens
z_t
CEM sampler
300 action sequences,
horizon 5 chunks
$a_{1..5}$
Predictor f_θ (frozen)
ViT, 3-frame history
imagines $\hat{z}_{1..5}$
$\hat{z}_{1..5}$
Trust scores u_k
MC-dropout, perturbations,
distance to data
repeat 30×
Trust gate
reject a plan if any
$u_k > \tau$; fallback ladder
accepted
Select and refit
lowest imagined cost;
refit to top 30
best plan
Execute
first 5 action chunks,
then observe again
actions
logged
u_k
EVALUATION: IMAGINED AGAINST REAL
Recorded plans
192 per start-goal pair,
from bad to good
Execute in simulator
every plan from the
same starting state
True error
imagined vs real, per step;
top 10% marked wrong
Results
AUROC of each score;
gate test